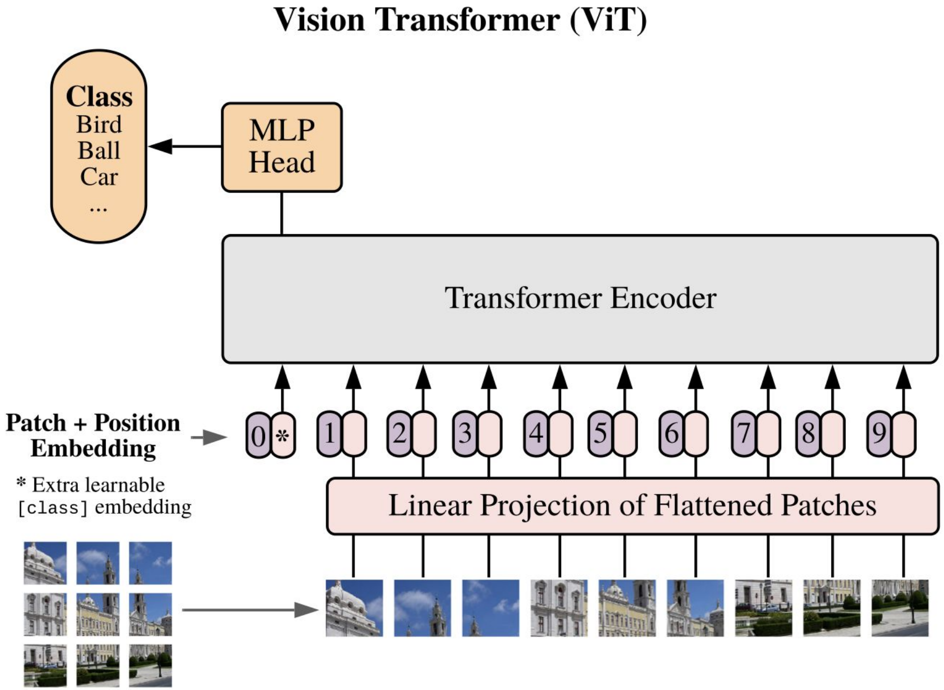
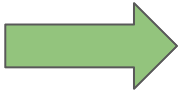
Transformers in Deep Learning

Lecture 5

Lecture 4 Recap

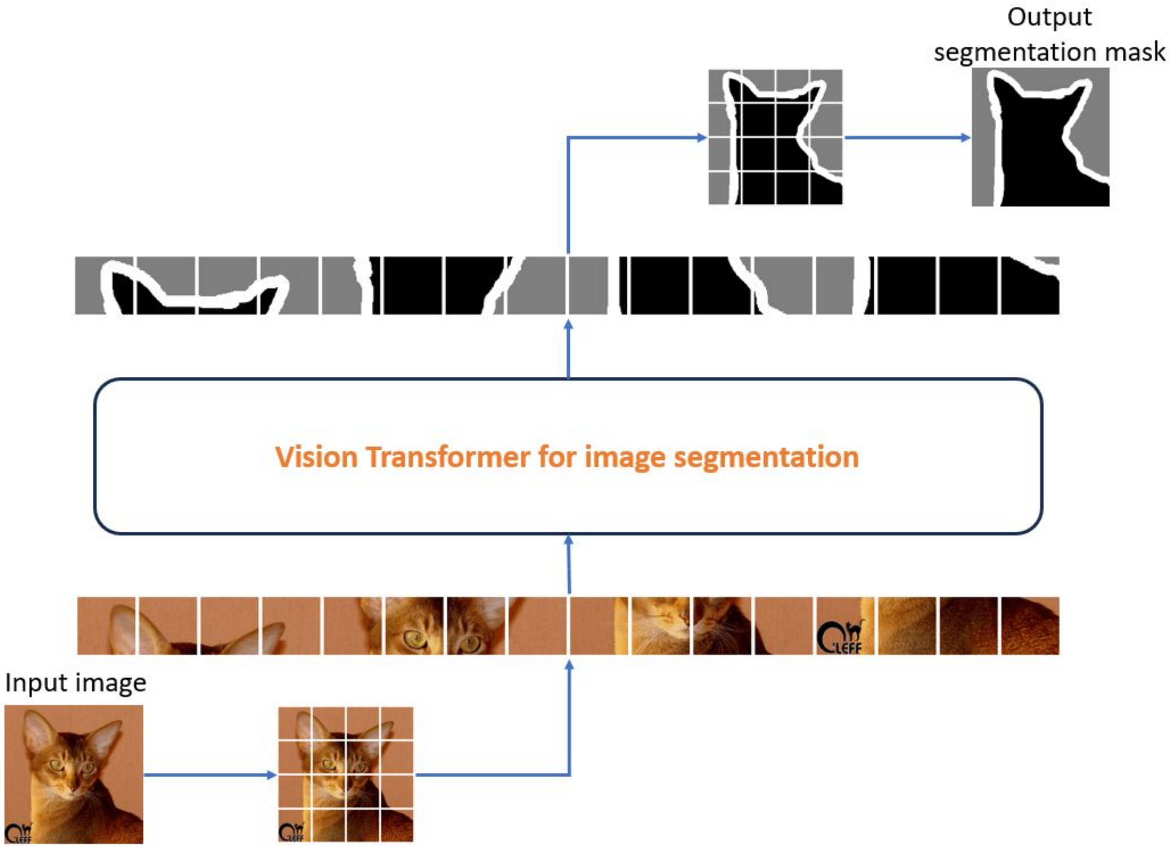




Classification

CAT

Lecture 4 Recap

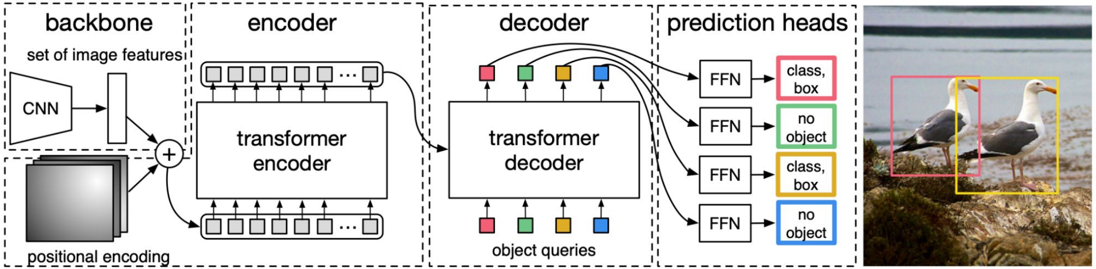


Semantic Segmentation

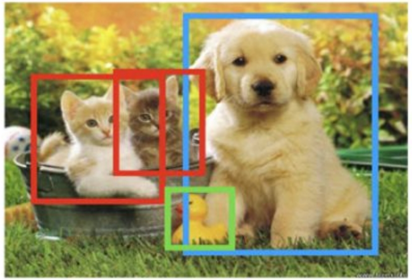
CAT GRASS
TREE

No object
Just pixels

Lecture 4 Recap



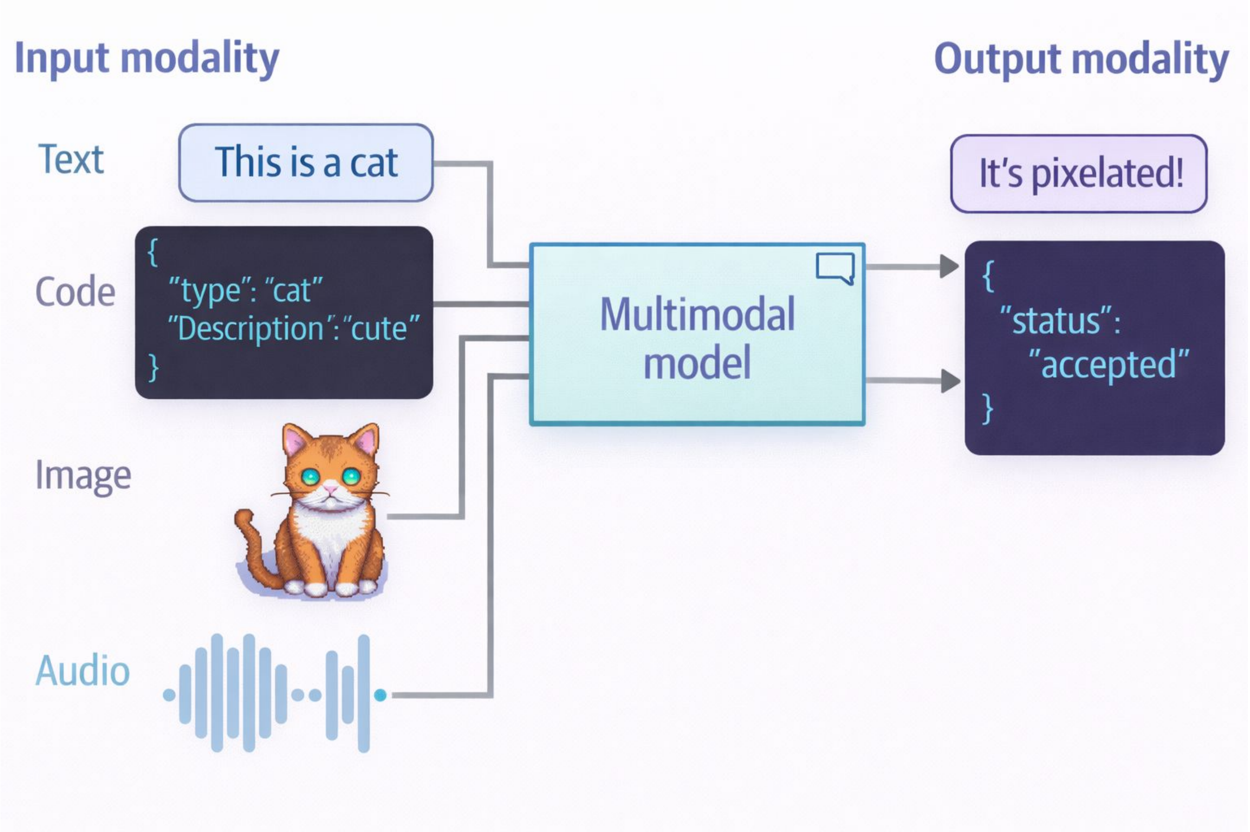
Object detection



CAT DOG DUCK

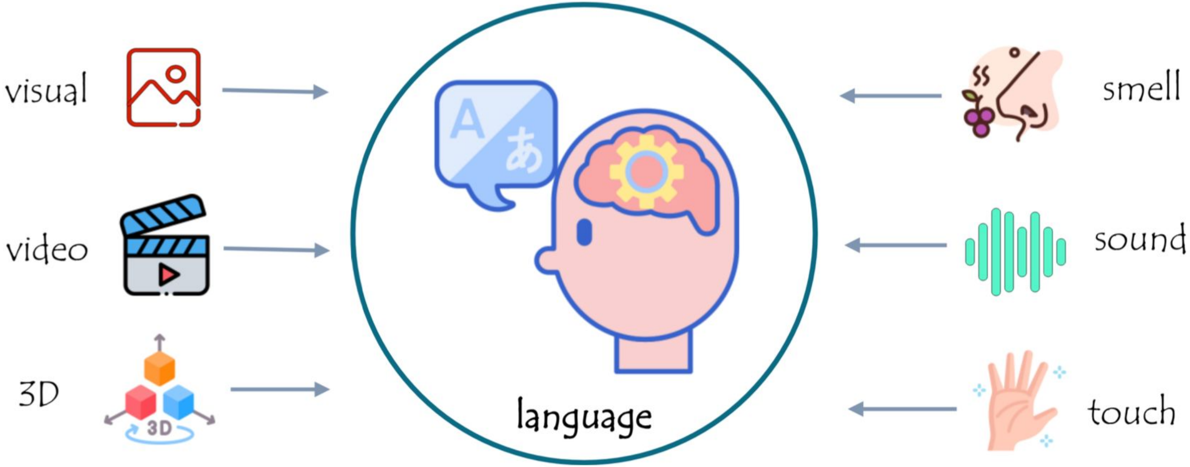
One hammer for everything:

Multimodality



Multimodality: key ideas

- Use world knowledge and reasoning from LLMs
- Explore various applications that emerge at the intersection of the modalities
- Boost quality of models using patterns in multimodal data that don't exist in unimodal data



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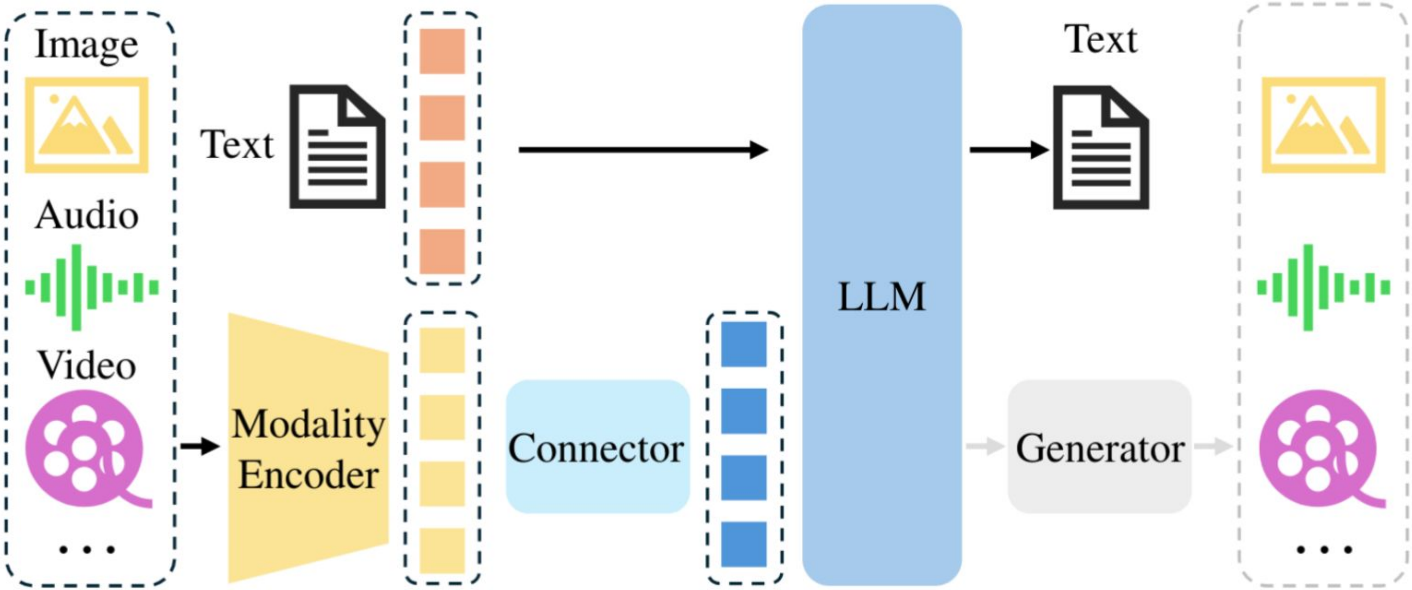
General scheme



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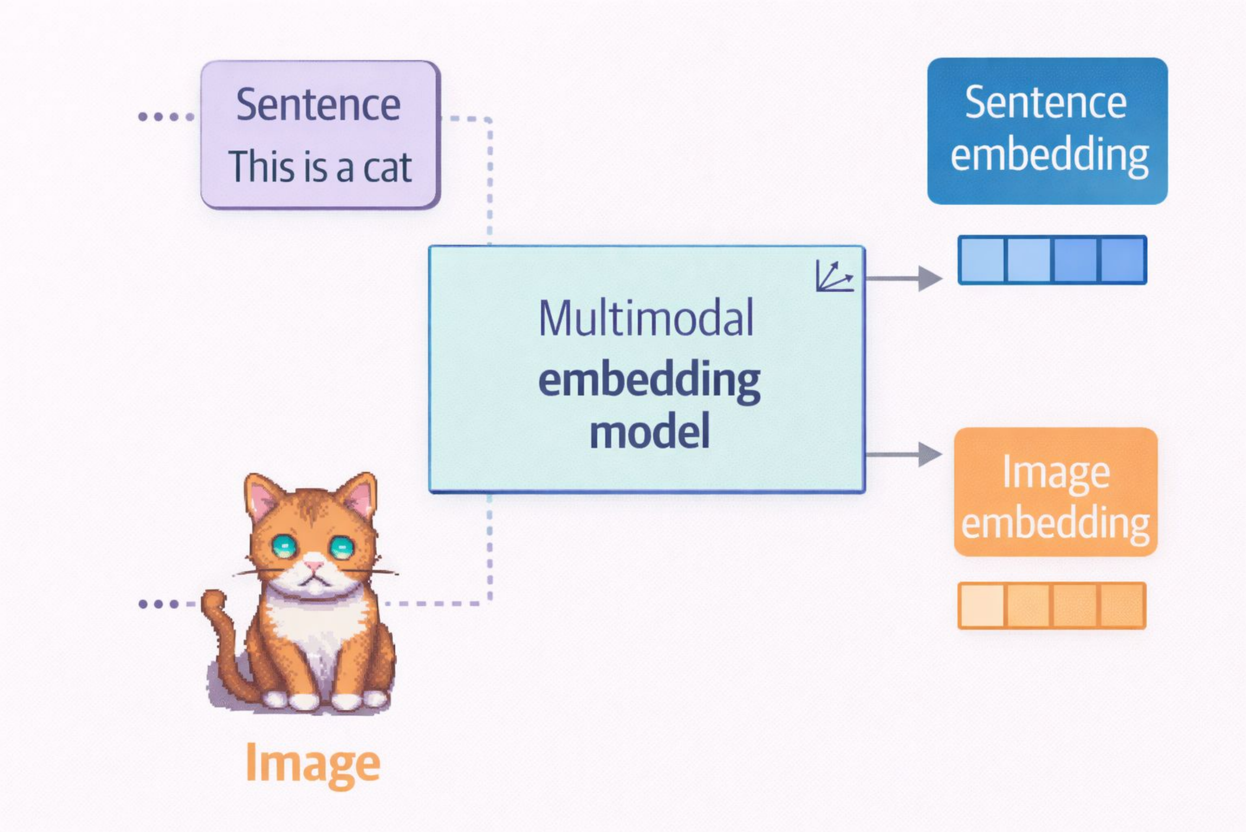
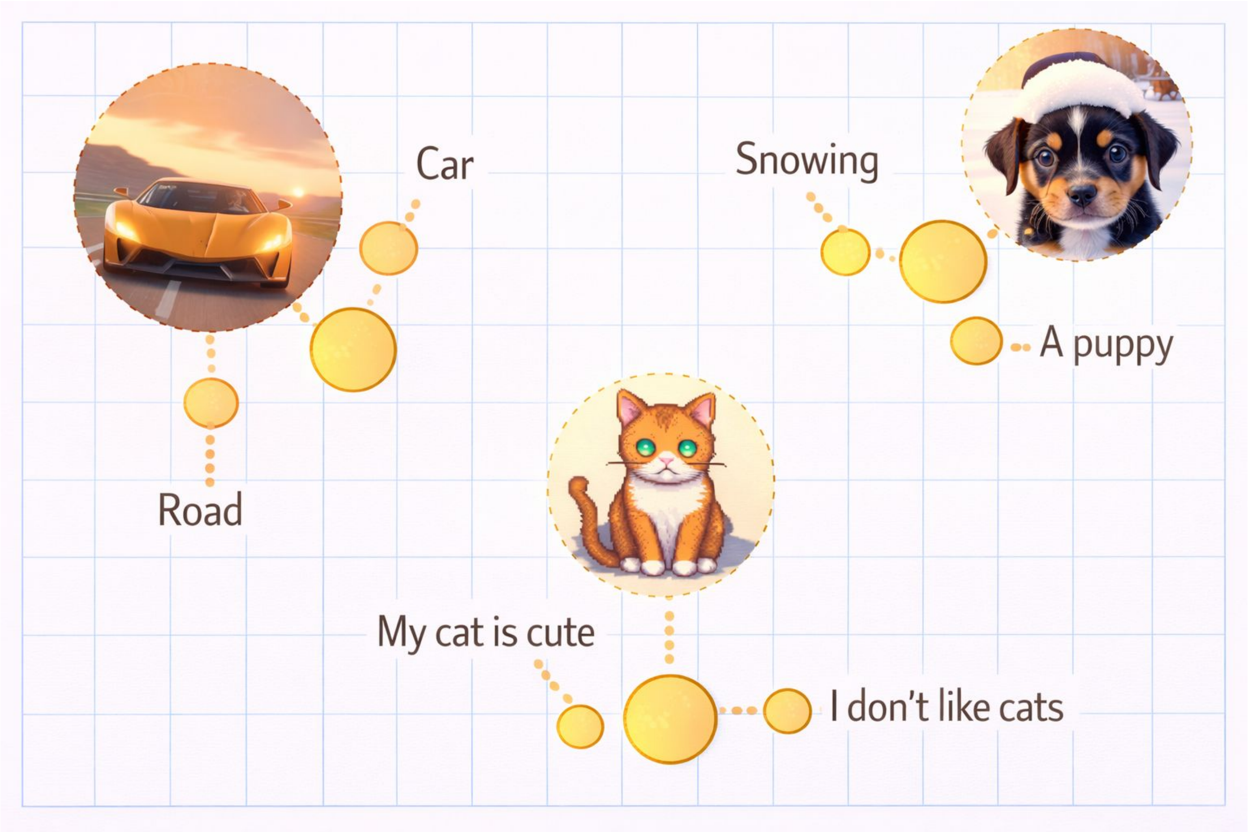
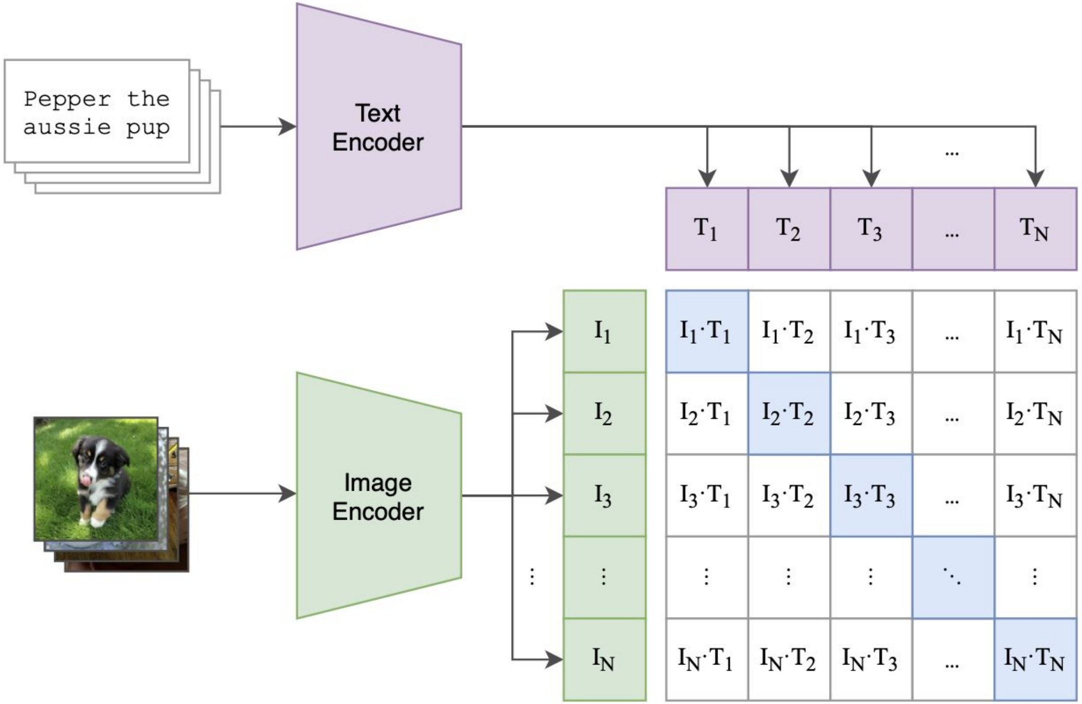
Multimodal embedding model



Multimodal embedding space



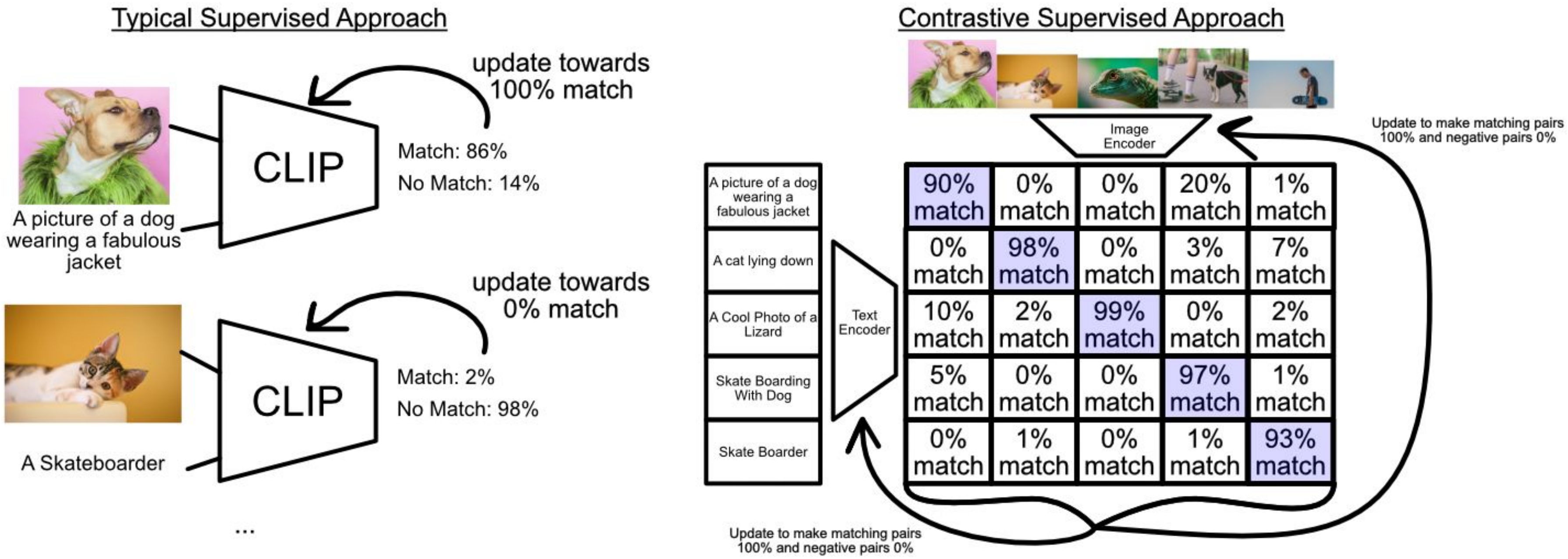
Multimodal embedding model: CLIP



400M (image, text) pairs, 500×V100 GPUs for pretraining

https://arxiv.org/abs/2103.00020

CLIP: Contrastive Learning



https://arxiv.org/abs/2103.00020

CLIP: Contrastive Loss

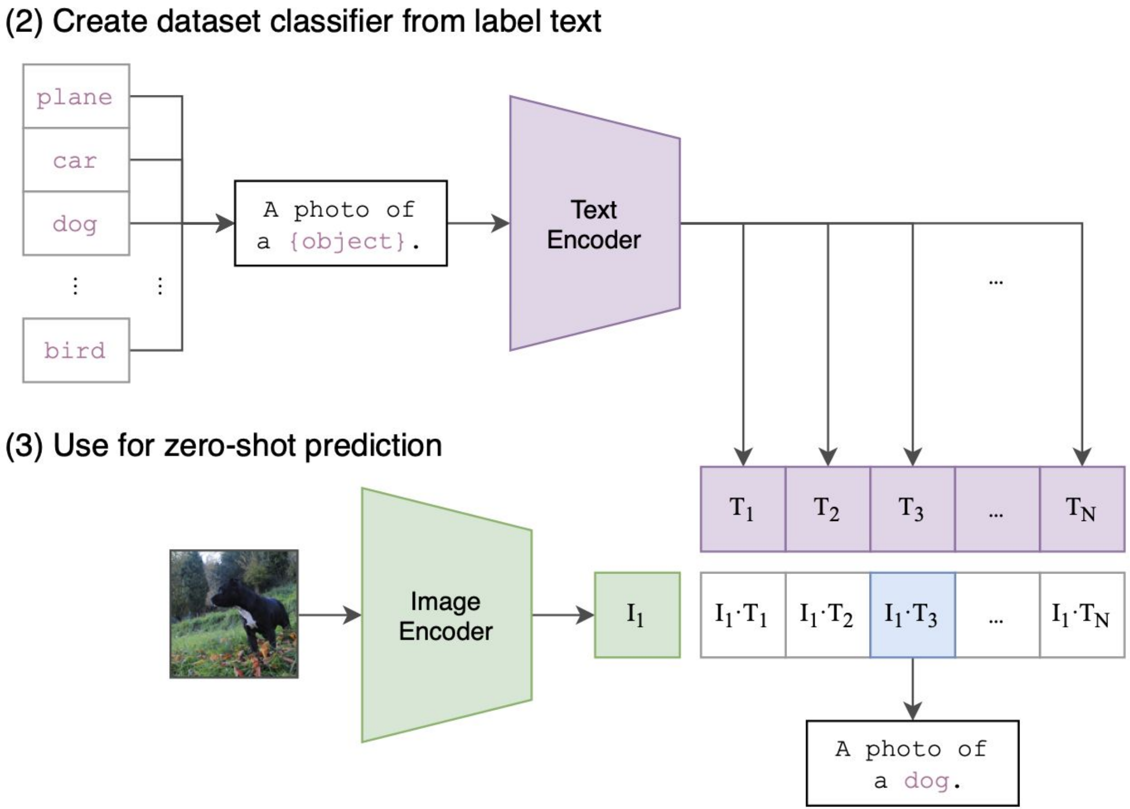
CLIP: Softmax

Image -> text softmax text -> image softmax

$$-\frac{1}{2N}\sum_{i=1}^{N}\left(\log\frac{e^{I_i \cdot T_i/\tau}}{\sum_{j=1}^{N} e^{I_i \cdot T_j/\tau}}+\log\frac{e^{I_i \cdot T_i/\tau}}{\sum_{j=1}^{N} e^{I_j \cdot T_i/\tau}}\right)$$

https://arxiv.org/abs/2103.00020

CLIP



https://arxiv.org/abs/2103.00020

SigLIP: better CLIP

CLIP: Softmax

Image -> text softmax   text -> image softmax

$$-\frac{1}{2N}\sum_{i=1}^{N}\left(\log\frac{e^{I_i\cdot T_i/\tau}}{\sum_{j=1}^{N} e^{I_i\cdot T_j/\tau}}+\log\frac{e^{I_i\cdot T_i/\tau}}{\sum_{j=1}^{N} e^{I_j\cdot T_i/\tau}}\right)$$

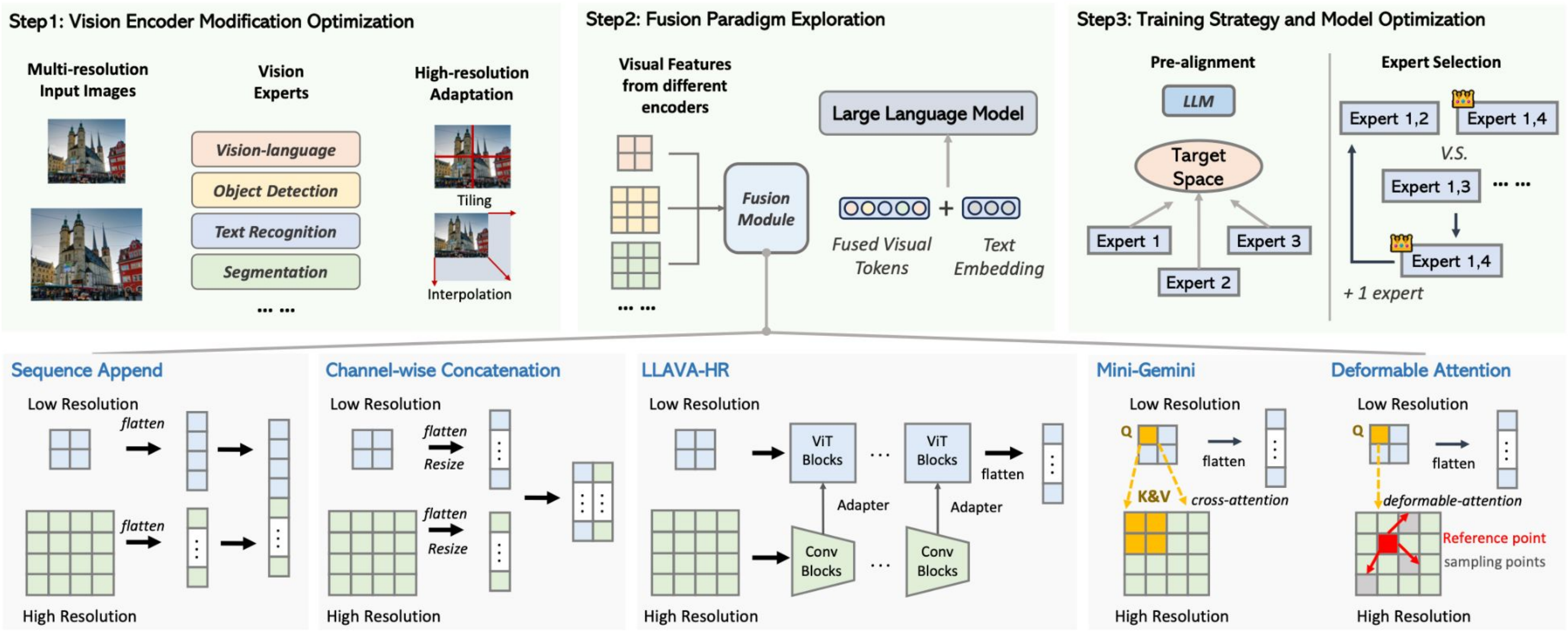
SigLIP: Sigmoid

$$-\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\log\frac{1}{1+e^{z_{ij}(I_i\cdot T_j/\tau+b)}}$$

- turn multiclass classification into binary classification of all pair (image, text) combinations
- no global normalization, hence better scaling and memory efficiency

https://arxiv.org/abs/2103.00020

Mixture of Encoders (Visual) (MOE)



https://arxiv.org/abs/2408.15998

High-level Classification

Multimodal models can be classified in 2 main types (4 subtypes)
based on the **fusion of input modalities**

1. Deep Fusion

deeply fuses multimodal inputs
within internal layers

- 1.1. **Standard** Cross-Attention (SC-DF)
- 1.2. **Custom** Layers (CL-DF)

2. Early Fusion

multimodal inputs are fed to the
model rather to its internals

- 2.1. **Non-tokenized** (NT-EF)
- 2.2. **Tokenized** (T-EF)

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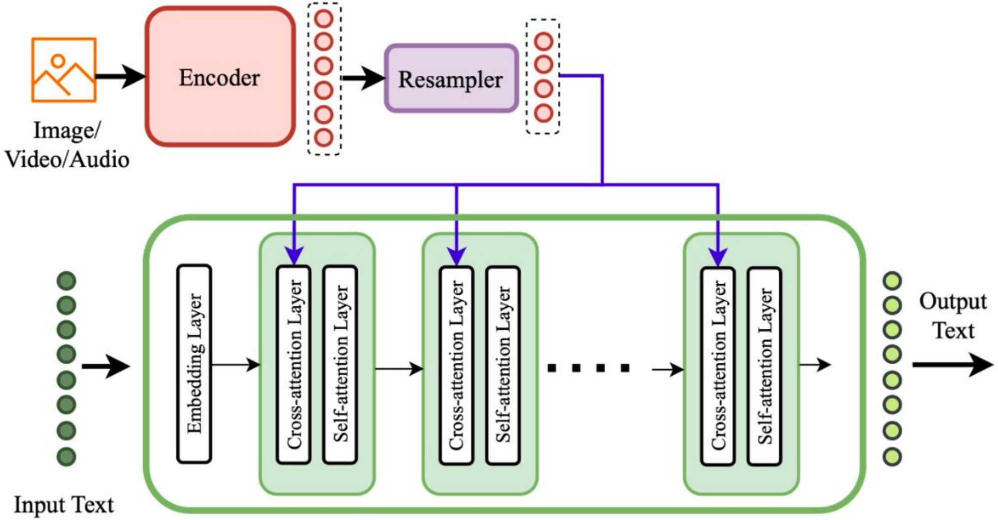
Deep Fusion:

Standard Cross-Attention Deep Fusion (SC-DF)

SC-DF: Standard Cross-Attention

Input modalities are deeply fused into the **internal layers of the LLM** using **standard cross-attention** layer



before

Flamingo

OpenFlamingo

Otter

Multimodal-GPT

after

VL-BART

VL-T5

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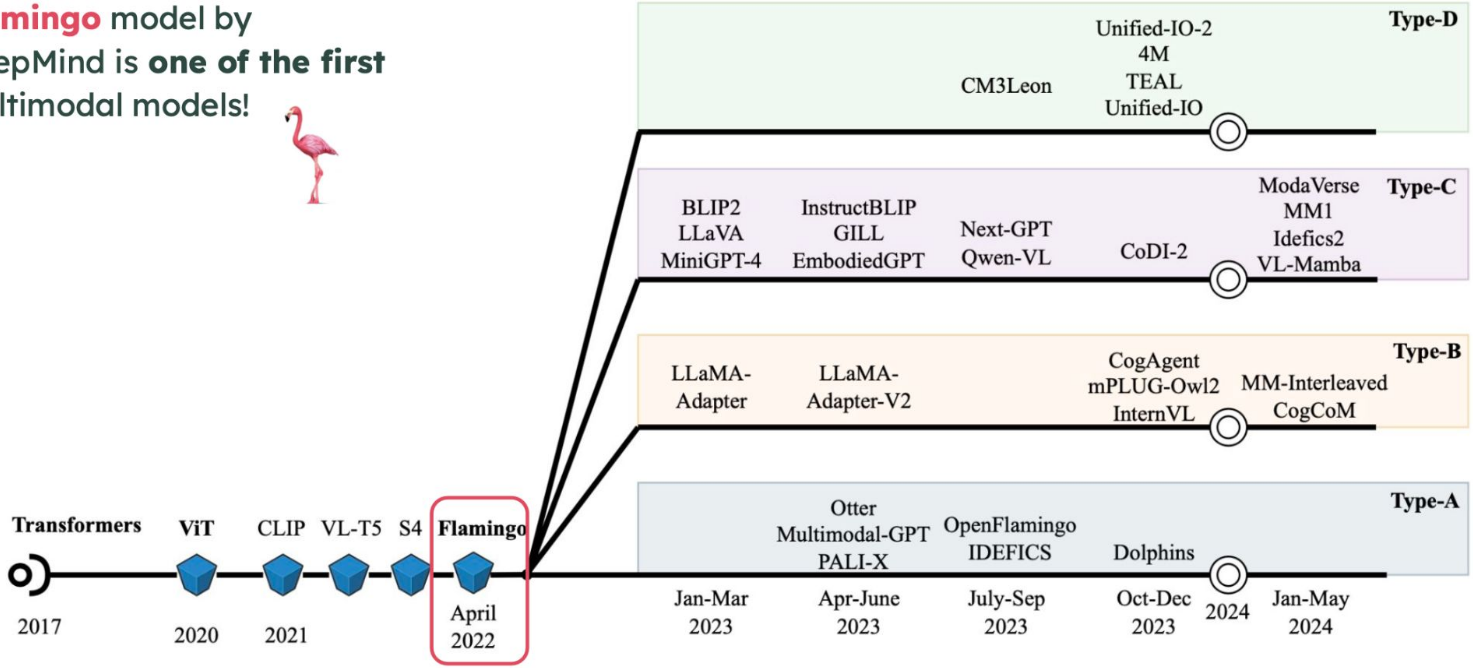
SC-DF: Standard Cross-Attention

Flamingo model by DeepMind is **one of the first** multimodal models!



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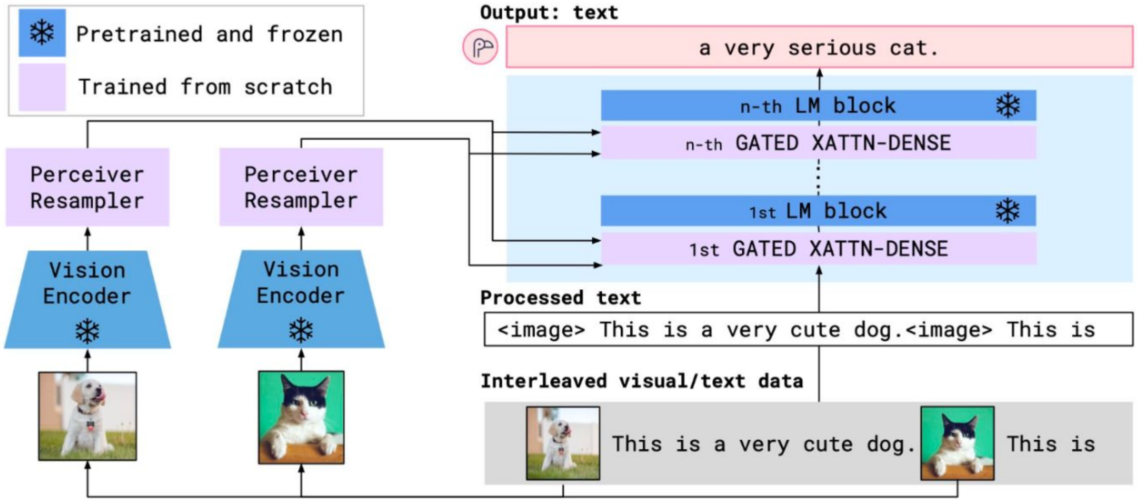
SC-DF: Open-Flamingo (nov 2022)



OpenFlamingo follows Flamingo architecture

vision model
CLIP ViT-L/14
(NFNet)

language model
RedPajama / MPT
(Chinchilla)

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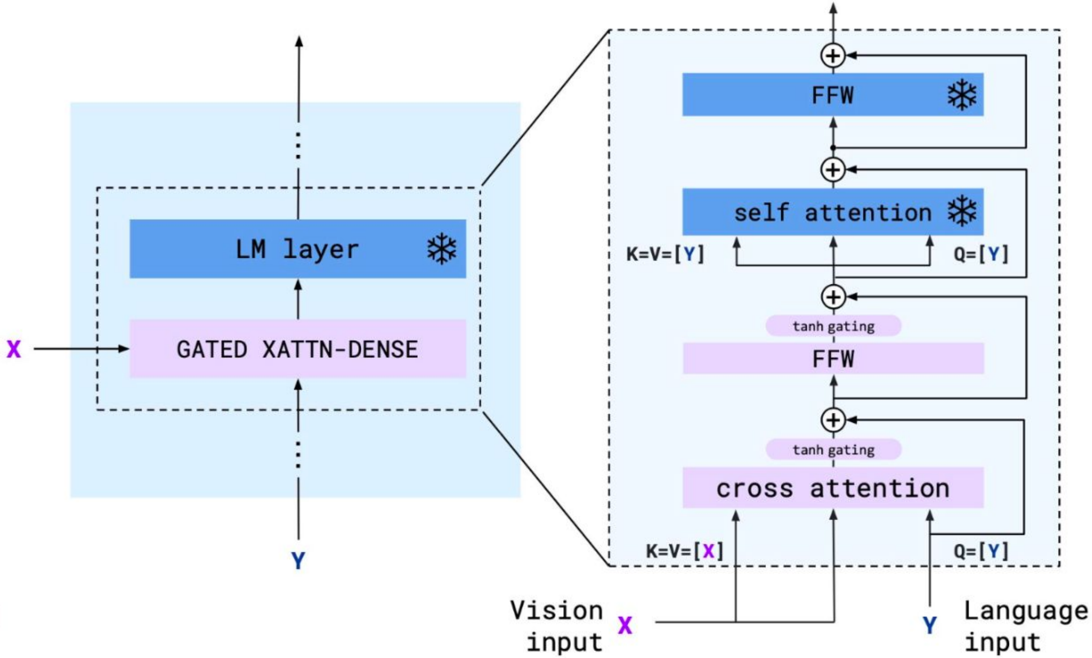
SC-DF: Open-Flamingo (nov 2022)

1. **freeze** the pretrained LM blocks
2. insert **gated cross-attention dense** blocks between the original layers
3. keep leyers gated to keep LM intact at initialization
4. **queries** = LM inputs

tanh-gating mechanism — multiples output of newly initialized layer by tanh(α)



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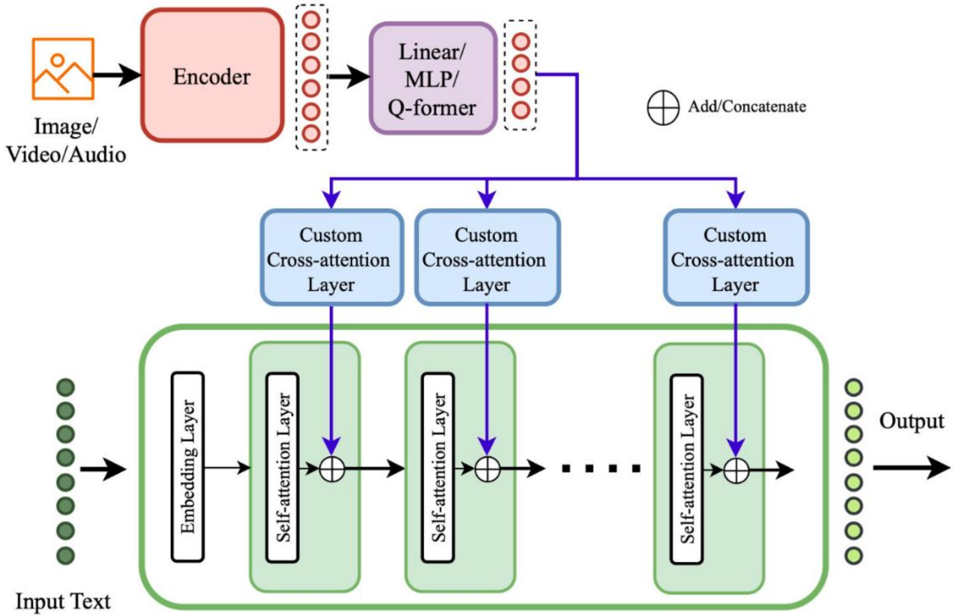
Deep Fusion:

Custom Layers Deep Fusion (CL-DF)

CL-DF: Custom Layers

Input modalities are deeply fused into the **internal layers of the LLM** using **custom-designed** layers



custom cross-attention

LLaMA-Adapter
CogVLM
InternVL
mPLUG-Owl2

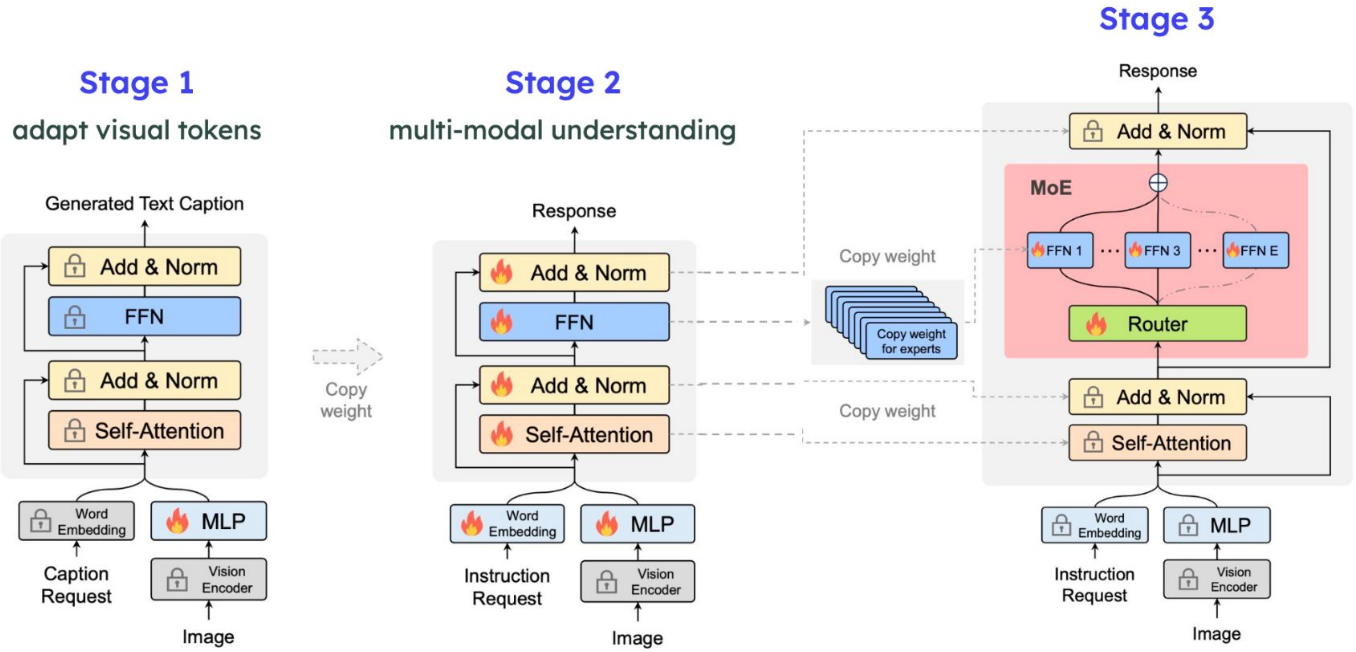
other custom layers

MoE-LLaVA
LION

https://code.mipt.ru/courses-public/cv/mm/public

CL-DF: MoE-LLaVA (dec 2024)

LLaVA – by Microsoft, **MoE-LLaVA** – Peking University: Mixture-of-Experts layer



many LLaMAs!

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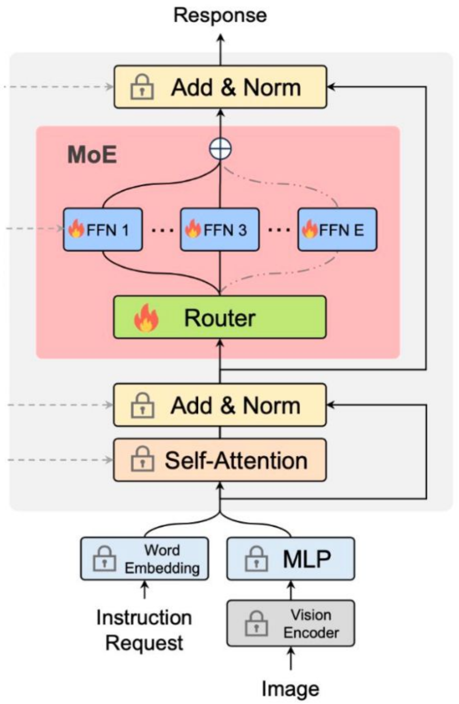
MoE-LLaVA: Router



1 have E experts, each expert = FFN

$$\mathcal{E} = [e_1, e_2, \cdots, e_E]$$

2 router = linear layer that assigns probabilities to experts

$$\mathcal{P}(\mathbf{x})_i = \frac{e^{f(\mathbf{x})_i}}{\sum_j^E e^{f(\mathbf{x})_j}}$$

3 calculate weighted sum

$$\mathrm{MoE}(\mathbf{x}) = \sum_{i=1}^{k} \mathcal{P}(\mathbf{x})_i \cdot \mathcal{E}(\mathbf{x})_i$$

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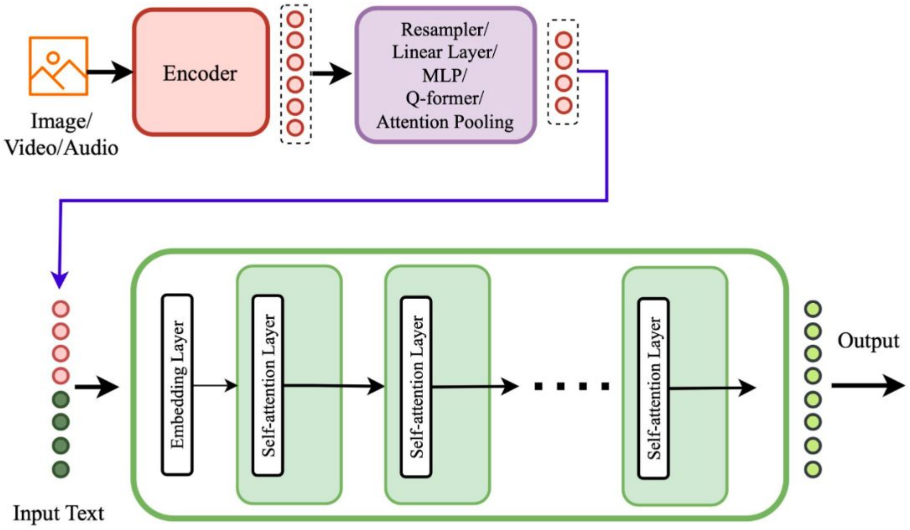
Early Fusion:

Non-Tokenized Early Fusion (NT-EF)

NT-EF: **Non-Tokenized**

Non-tokenized input modalities are **directly fed to the model** rather than to internal layers



- **Q-Former:** BLIP-2 🥇, MiniGPT-v2
- **Custom layer:** **Qwen-VL**, AnyMAL, Video-ChatGPT, EmbodiedGPT
- **Linear / MLP:** DeepSeek-VL, LLaVA, LLaVA-NeXT, PaLM-E, Shikra
- **Perceiver resampler:** Monkey, V*, Kosmos-G

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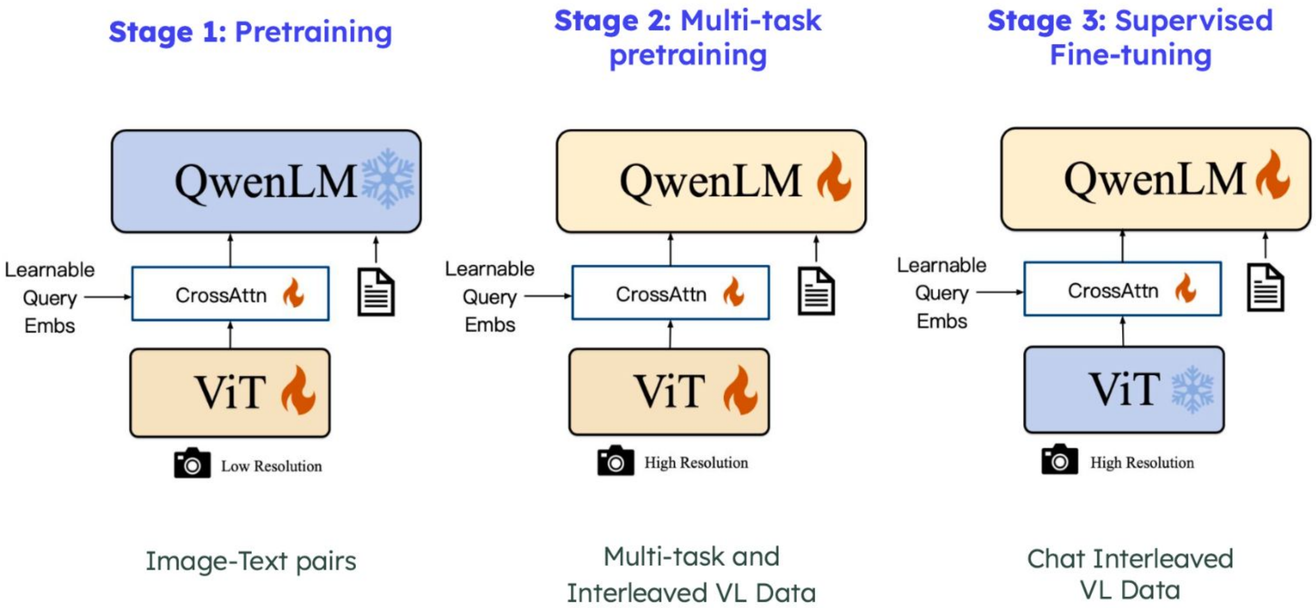
NT-EF: **Qwen-VL** (oct 2023)



vision model
OpenClip ViT-bigG

language model
Qwen-7B

https://code.mipt.ru/courses-public/cv/mm/public

Early Fusion:

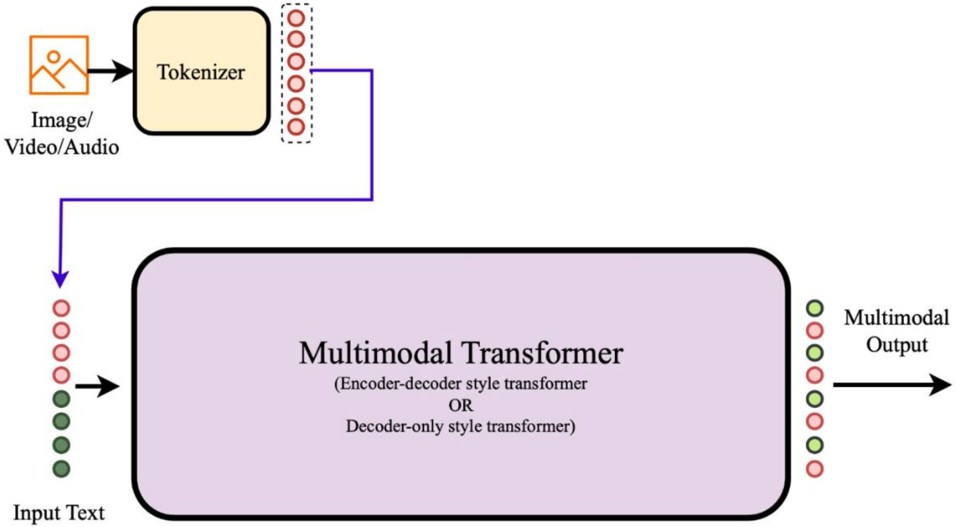
Tokenized Early Fusion (T-EF)

https://code.mipt.ru/courses-public/cv/mm/public

T-EF: **Tokenized**

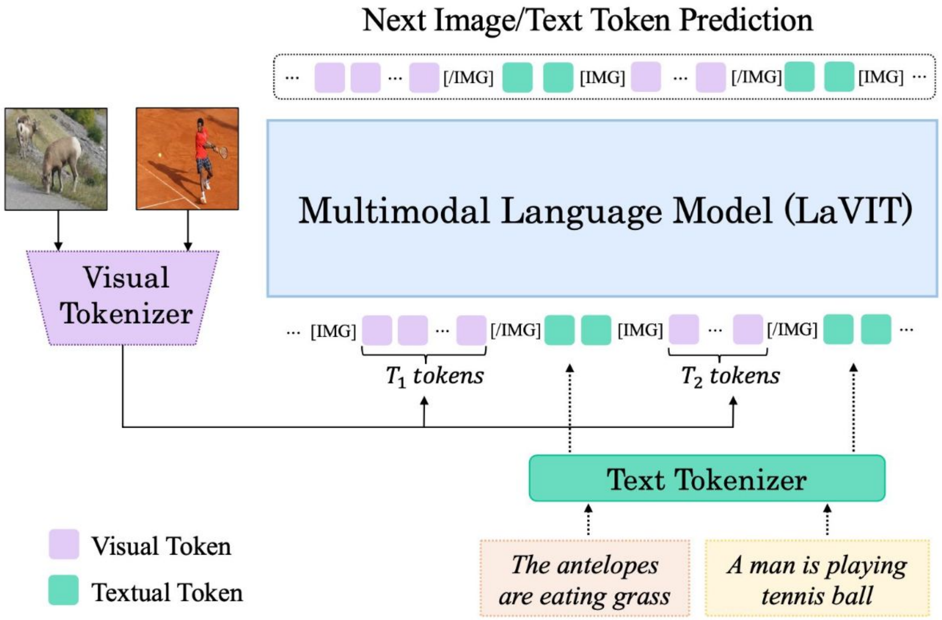
Inputs are tokenized **using a common tokenizer** or modality specific tokenizers

decoder-only transformer	**encoder-decoder transformer**
LaVIT	Unified-IO
TEAL	Unified-IO
CM3Leon	4M
VL-GPT	



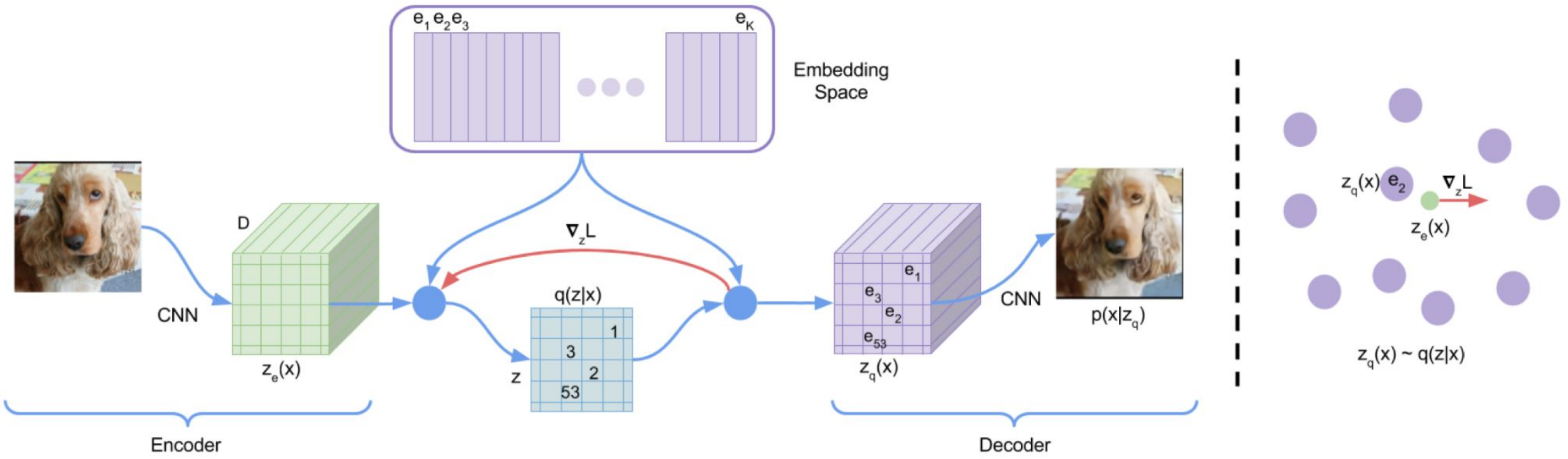
https://code.mipt.ru/courses-public/cv/mm/public

T-EF: **LaVIT** (mar 2024)



https://code.mipt.ru/courses-public/cv/mm/public

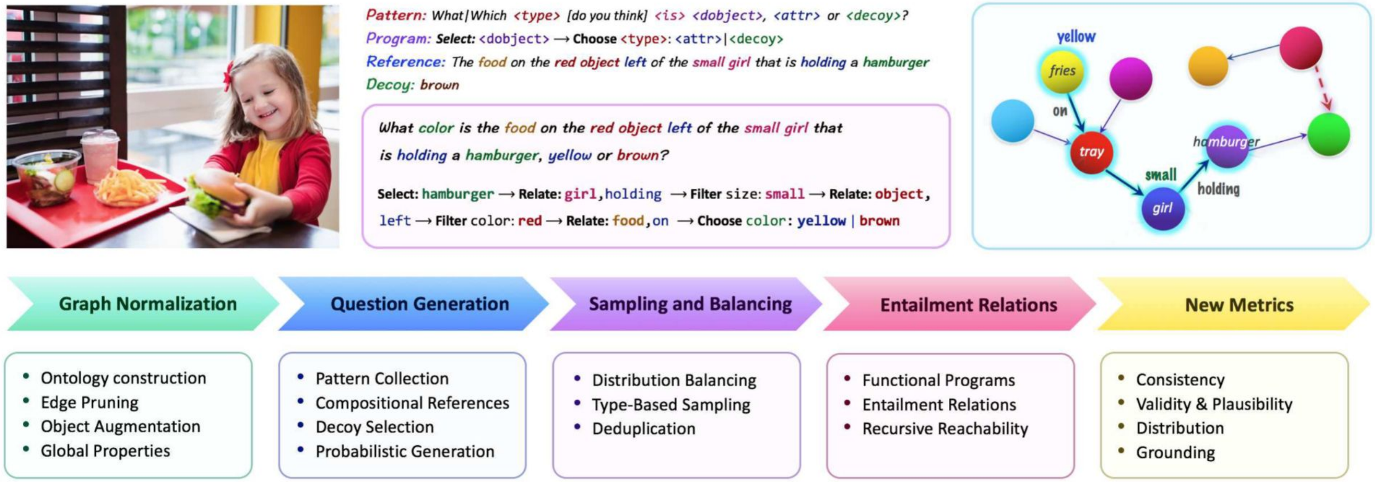
Image tokenization: VQ-VAE



https://arxiv.org/abs/1711.00937

Benchmarks

Static benchmarks: GQA



Hudson et al. GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering. CVPR 2019

https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Static benchmarks: GQA

A1. Is the **tray** on top of the **table** black or light brown? light brown
A2. Are the **napkin** and the **cup** the same color? yes
A3. Is the small **table** both oval and wooden? yes
A4. Is there any **fruit** to the left of the **tray** the **cup** is on top of? yes
A5. Are there any **cups** to the left of the **tray** on top of the **table**? no
B1. What is the brown **animal** sitting inside of? **box**
B2. What is the large **container** made of? cardboard
B3. What **animal** is in the **box**? **bear**
B4. Is there a **bag** to the right of the green **door**? no
B5. Is there a **box** inside the plastic **bag**? no

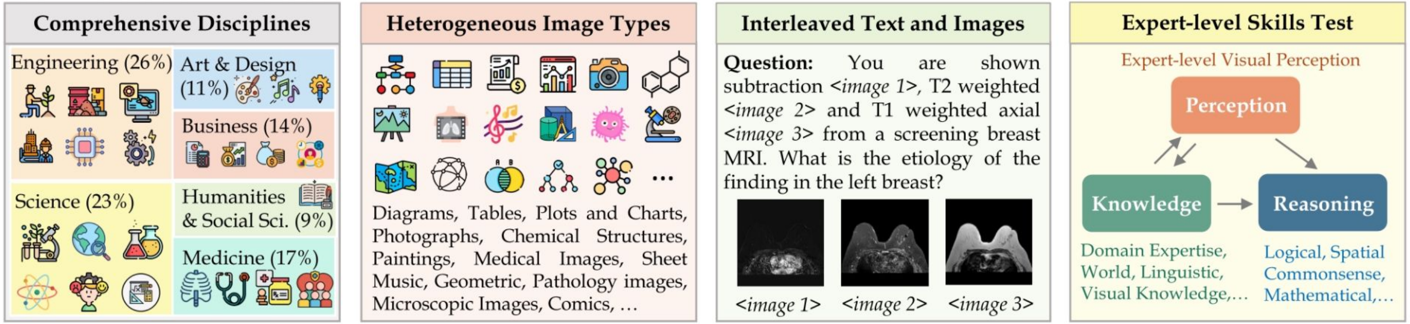
- questions generated using scene graph of images
- 22.6M questions for 113k images
- evaluation metric: accuracy along with 5 more detailed metrics

https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Static benchmarks: MMMU

- 11.5k questions from 6 university disciplines
- answers are extracted using regexps
- evaluation metric: accuracy

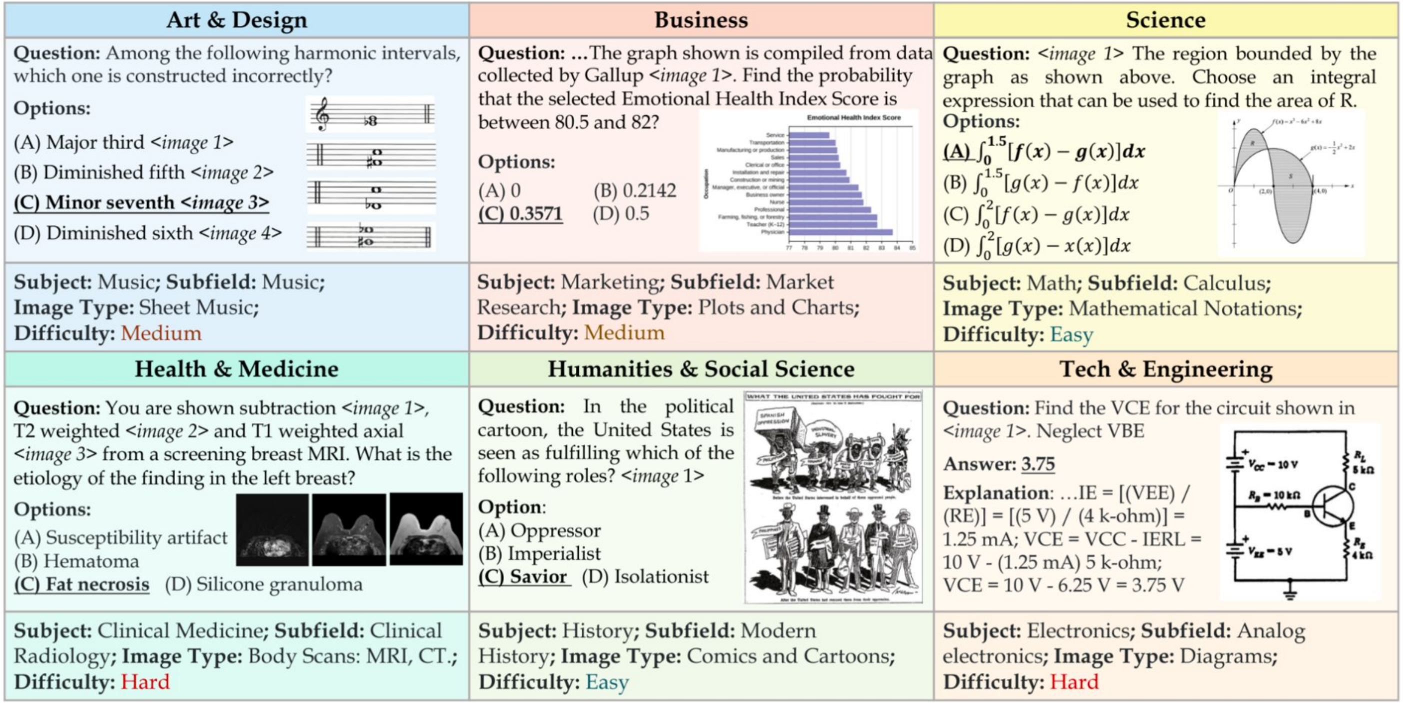


Yue et al. MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. CVPR 2024

https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Static benchmarks: MMMU



https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Static benchmarks: TextVQA

Singh et al. Towards VQA Models That Can Read. CVPR 2019

- 45k questions for 28k images, 10 answers per question
- evaluation metric: VQA accuracy (100% correct if 3 humans provided that answer)

https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Statistic	Number
Total Votes	8,076
Anonymous	6,636
Non-anonymous	1,440
Left Vote	2,932
Right Vote	2,839
Tie Vote	979
Bad Vote	1,326
Days	102
Total Round	10,884
Avg Round	1.34
Avg Token Input	31.00
Avg Token Output	108.87

Table 1: Statistics of votings in WV-ARENA.

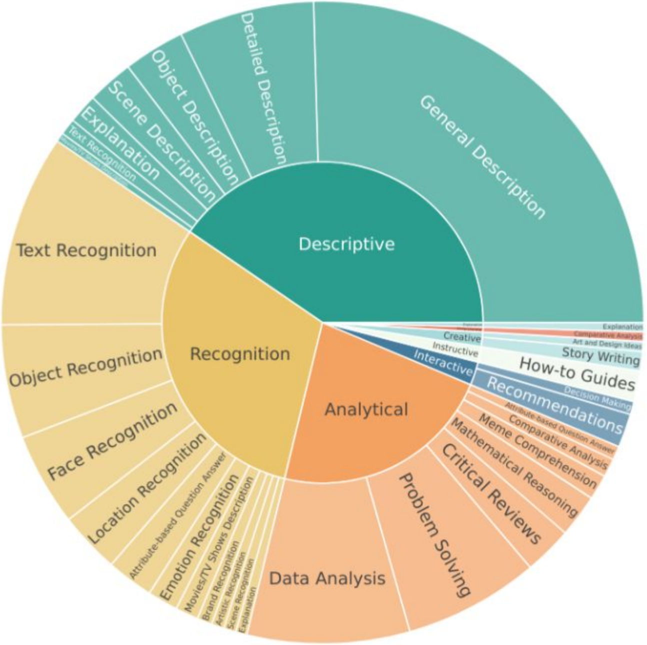


Figure 2: Question Category

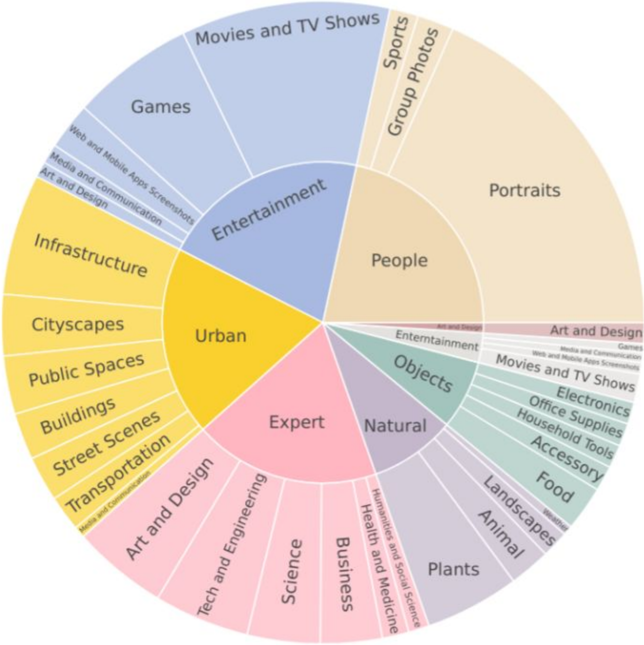


Figure 3: Image Domain

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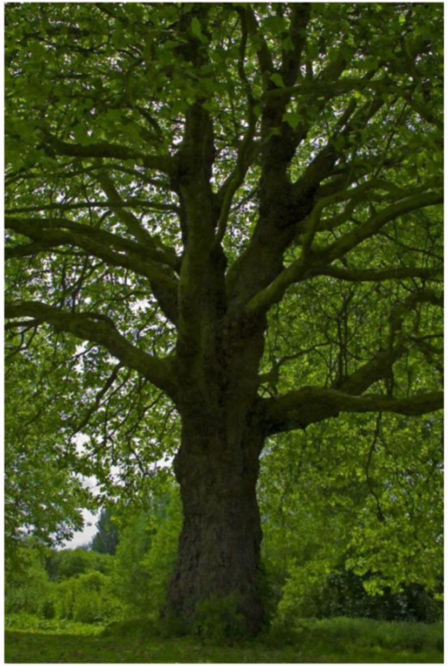
Benchmarks

Image [Entertainment-Movies/TV Shows]

[Descriptive-Movies/TV Shows] **Text Prompt:** What are the two giraffe characters on this movie poster doing?

Image [Natural-Plants]



[Analytical-Problem Solving] **Text Prompt:** How likely is it to snow after this picture was taken? What would change with this type of tree before it's likely to snow?

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Benchmarks

Image [Urban-Buildings]

[Recognition-Location] **Text Prompt:** where is this?

Image [Expert-Science]

[Analytical-Safety Procedures] **Text Prompt:** Can you tell me the potential risks and the unreasonale parts in the image?

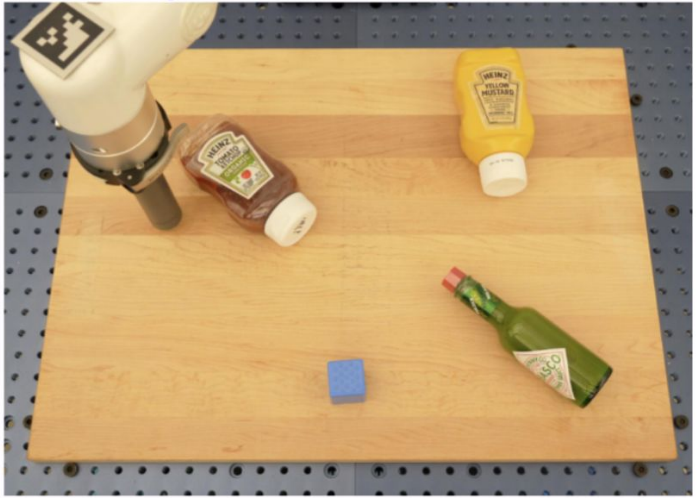
https://code.mipt.ru/courses-public/cv/mm/public

Benchmarks

Image [Natural-Landscapes]

[Recognition-Location] **Text Prompt:** where was this photo taken?

Image [Objects-Household Tools]



[Descriptive-Object Description] **Text Prompt:** describe the scene and objects

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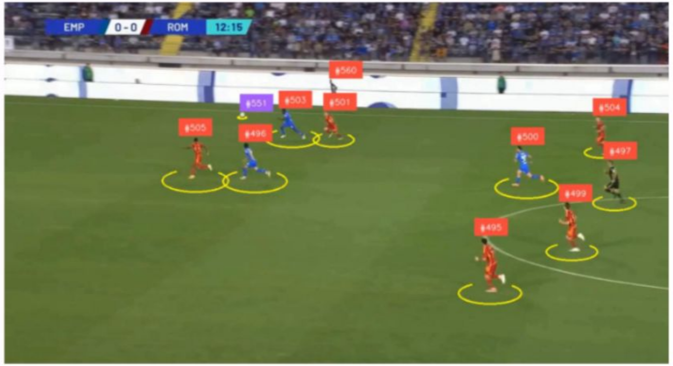
Benchmarks

Image [Entertainment-Web and Mobile Apps Screenshots]

[Interactive-Web Navigation] **Text Prompt:** I need to download flyer, you will be given screenshot from browser with elements marked with number. give next action to take on web page to download the flyersngive me response in below format example 1 action:[click,scroll,wait], box:1 format action:, box:

Image [Event-Sports]



[Descriptive-Scene Description] **Text Prompt:** this is a football match , every player has an identifier , describe every player action (example : player #501 is running)

https://code.mipt.ru/courses-public/cv/mm/public